Chunks, Tokens and Embeddings

A very simple step-by-step example for understanding the RAG data flow

Example document

Python is a programming language. It is an object-oriented programming language. FastAPI creates web APIs.

1. Document

The complete text or file is called a **document**. A document may be a PDF, DOCX, TXT file, webpage or any other source of text.

```
Python is a programming language.
It is an object-oriented programming language.
FastAPI creates web APIs.
```

2. Chunks

The document is divided into smaller parts called **chunks**. Smaller chunks make searching and processing easier.

```
Chunk 1: Python is a programming language.

Chunk 2: It is an object-oriented programming language.

Chunk 3: FastAPI creates web APIs.
```

3. Tokens

Each chunk is broken into small text units called **tokens**. A token may be a word, part of a word or punctuation. The exact split depends on the model.

```
Chunk 1 tokens:
["Python", "is", "a", "programming", "language", "."]

Chunk 2 tokens:
["It", "is", "an", "object", "-", "oriented", "programming", "language", "."]

Chunk 3 tokens:
["FastAPI", "creates", "web", "APIs", "."]
```

4. Embedding Model

The **embedding model** reads each chunk and represents its meaning. It understands that the first two chunks discuss programming, while the third discusses FastAPI and web APIs.

```
Chunk 1 -> Embedding Model
Chunk 2 -> Embedding Model
Chunk 3 -> Embedding Model
```

5. Numerical Vectors

The embedding model converts the meaning of every chunk into a list of numbers called a **vector**. The numbers below are simplified examples; real vectors usually contain hundreds or thousands of values.

```
Chunk 1 -> [0.12, 0.81, -0.34, 0.56, ...]

Chunk 2 -> [0.15, 0.76, -0.29, 0.61, ...]

Chunk 3 -> [0.89, -0.21, 0.48, 0.17, ...]
```

Important: Chunk 1 and Chunk 2 may have similar vectors because they have related meanings. Embeddings store **meaning**, not just exact words.

6. Vector Database

The vector database stores each original chunk together with its numerical vector. Examples include ChromaDB, FAISS, Pinecone and PGVector.

Chunk	Original text	Example vector
1	Python is a programming language.	[0.12, 0.81, ...]
2	It is an object-oriented programming language.	[0.15, 0.76, ...]
3	FastAPI creates web APIs.	[0.89, -0.21, ...]

When a User Asks a Question

Suppose the user asks:

Which tool creates web APIs?

The same embedding model converts the question into a vector:

```
Question -> [0.86, -0.19, 0.51, 0.20, ...]
```

The vector database compares the question vector with stored vectors and finds the most similar chunk:

Most similar result:

Chunk 3: FastAPI creates web APIs.

Complete Process

DOCUMENT	Complete text
↓	
CHUNKS	Smaller parts of the document
↓	
TOKENS	Words, word pieces and punctuation
↓	
EMBEDDING MODEL	Understands and converts meaning
↓	
VECTORS	Meaning represented as numbers
↓	
VECTOR DATABASE	Stores and searches vectors
↓	
RELEVANT RESULT	FastAPI creates web APIs

Easy memory formula:

Document = complete material | Chunk = small paragraph | Token = small text unit | Embedding = numerical meaning